Power Budget

Team Number:	Team 205
Project Name:	**Project B.A.M.F**
Team Member Name:	**Herman Williams**

All Major Components	**Component Name**	**Part Number**	**Supply**	**#**	**Absolute**	**Total**	**Unit**
	Solenoid	1528-2797-ND	+5 - 6V	1	800	800	mA
	DC motor	942-PPN7PA1	+5 V	2	840	1680	mA
	Curiosity nano	PIC18F57Q43	1.8-5.1V	1	500	500	mA
+5V Power Rail	**Component Name**	**Part Number**	**Supply**	**#**	**Absolute**	**Total**	**Unit**
	Solenoid	1528-2797-ND	+5 - 6V	1	800	800	mA
	DC motor	942-PPN7PA1	+5 V	2	840	1680	mA
	Curiosity nano	PIC18F57Q43	1.8-5.1V	1	500	500	mA
					Subtotal	2980	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	3725	mA
Regulator	+5V Regulator	LM7805	0-35 V	1	1500	1500	mA
					Total Remaining Current Available on +5V Rail	-2225	mA

External Power Source 1	**Component Name**	**Part Number**	**Supply**	**Output**	**Absolute**	**Total**	**Unit**
Power Source 1 Selection	Plug-in Wall Supply	S39-2402500-D1	110VAC	+12V	2500	2500	mA
Power Rails Connected to External Power Source 1							
	+5V Regulator	LM7805	0-35V	1	1500	1500	mA
					Total Remaining Current Available on External Power Source 1	1000	mA